Portable Coordinate Measuring Arm Joint Design

Mechatronic Project 478
Final Report

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Executive summary

Title of Project
Joint Design for a Portable Coordinate Measuring Arm (PCMA)
Objectives
To derive technical performance measures (TPMs) from an error-propagation model, and use them to design, build, and test a position-holding joint that maintains and /or improves the arm's overall accuracy.
What is current practice and what are its limitations?
Current joints rely on counterbalancing and lack holding torque, meaning the operator must physically support the arm's weight during operation.
What is new in this project?
The introduction of a position-holding mechanism specifically engineered using mathematically derived TPMs.
If the project is successful, how will it make a difference?
It will directly maintain or improve the physical measurement accuracy of the arm. Additionally, it will yield a modified kinematic model to serve as a generalized framework for future accuracy analysis.
What are the risks to the project being a success? Why is it expected to be successful?
Risks: University workshop manufacturing capabilities may fall short of required tolerances, and the holding mechanism could introduce unwanted friction. Expectation of Success: The design requirements are heavily grounded in an established, proven error model.
What contributions have/will other students made/make?
The project builds directly upon an initial error-propagation model developed by Stellenbosch alumna Leon Hall.
Which aspects of the project will carry on after completion and why?
The modified error-propagation model. It will be able to accept custom arm dimensions and compute the uncertainty distribution within a given work volume, forming the mathematical foundation for future PCMA development.
What arrangements have been/will be made to expedite continuation?

Extensive, standardized documentation will be left behind to ensure future researchers can build directly onto this work without repeating groundwork.

ECSA Exit Level Outcomes

This section addresses the following Engineering Council of South Africa (ECSA) exit level outcomes and indicates how each outcome was demonstrated in the project.

GA 1 - Problem-Solving

The project addresses an engineering problem, which is the lack of a joint design for PCMAs that meet accuracy requirements that account for spatial dependency and proper position-holding implementation without the impedance of uninhibited movement.

GA 2 - Application Scientific and Engineering Knowledge

GA 3 - Engineering Design

GA 5 - Use of Engineering Tools

GA 6 - Professional and Technical Communication

GA 8 - Individual and Collaborative Teamwork

GA 9 - Independent Learning Ability

Acknowledgements

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Constants

$L_0 =$	300 mm

Variables

Re_{D}	Reynolds number (diameter)	[ ]
x	Coordinate	[m]
$\ddot{x}$	Acceleration	[m/s^2]
θ	Rotation angle	[rad]
τ	Moment	[N·m]

Vectors and Tensors

$\vec{v}$	Physical vector, see equation ...

Subscripts

a	Adiabatic
a	Coordinate

Abreviations

3D	three-dimensional
AI	artificial intelligence
CAD	computer-aided design
D-H	Denavit-Hartenberg
FEA	Finite Element Analysis
PCMA	portable coordinate measuring arm
SPAT	single-point articulation test

Chapter 1

Introduction

1.1 Background

A portable coordinate measuring arm (PCMA) is a handheld metrology instrument that captures three-dimensional (3D) coordinates directly from a physical object, as shown in 1.1 below, (1).

Figure 1.1: A portable coordinate measurement arm, (1)

PCMAs offer adaptable measurement systems for multiple applications, such as dimensional analysis, computer-aided design (CAD) inspection, reverse engineering, etc., (2). The main limitation pf PCMAs is the effect of manual operation, which can introduce positional instability and dynamic structural deformation during measurement, (3). Manual operation can also inhibit calibration procedures as it is required that the PCMA must be held still during the procedure, (4).
As a result, designing an effective joint with a position-holding mechanism would improve the design of a PCMA and a good practice of safety. By designing the joint in this manner, error due to manual operation will be minimised, kinematic stability and accuracy will be improved.

1.2 Aim and Objectives

The aim of this project is to implement a position-holding mechanism on a PCMA without jeopardising its accuracy and ease of movement.
The objectives of the project are:

1. To generalise the conventional Denavit-Hartenberg(D-H) model derived by Hall and Schreve by sampling the working volume of a PCMA determining distribution of uncertainty along its length, (5).

2. To design, build, and test the joint design through the single-point articulation test (SPAT), such that it maintains and/or improves the accuracy of the PCMA.

3. To compare the results from SPAT to the D-H model and verify the joint design's effect on the PCMA's accuracy.

1.3 Scope

In the process of determining D-H parameters, operation manual and available schematics will be used. In addition, only joint will be designed, built, and tested for data acquisition, through calibration and accuracy testing via SPAT.

1.4 Motivation

PCMAs are becoming more popular than tradition coordinate measuring machines (CMM) for in-line[1] and on-site industrial measurement, (1). They have broad applicability across multiple industries such as aerospace, automotive, heavy manufacturing, where having a portable measurement instrument decreases handling time, production delays, and revision of work.
Despite their growing popularity, the PCMA's revolute joints must satisfy competing demands, including:

- Tight run-out tolerances for spatial accuracy.

- Sufficient holding torque to keep the arm stationary once released.

- Internal cable routing.

This has proved difficult since there are found commercial available joints that meet these demands. In addition, it has been established that angular position errors at each joint propagate through the kinematic chain, which affects end-effector's

[1]In the context of PCMAs, ”in-line” refers to performing quality inspections and measurements directly on the production shop floor, instead of a temperature-controlled quality laboratory, (1).

accuracy, proving that joints have a measurable impact on the PCMA's accuracy, (6). This project addresses the gap by deriving a joint specifications from a generalised model based off of an error model that was developed by a Stellenbosch Alumnus, Leon Hall, (5). This will produce a validated joint and testing methodology to support future PCMA development.

1.5 Use of AI Tools

Gemini is an artificial intelligence (AI) tool will be used check grammatical errors and sentence structure as well as LaTex formatting for the report.

1.6 Report Overview

This section follows the following structure:

- Section 1 introduces the project background, aim and objectives, scope, and motivation for project.
- Section 2 presents a literature review discussing sources relevant to PCMA joint design, position-holding technology, kinematic modeling, error analysis, and PCMA testing.
- Section 3 describes the derivation of specifications for the joint design.
- Section 4 details concept development and evaluation.
- etc. etc.

Chapter 2

Literature review

2.1 PCMA Context

Sources such as Leussink Engineering and Meagher gave context as what PCMAs are and how they are used.

2.2 The Denavit-Hartenberg Model

Chapter 3

Content chapter

Unless the chapter heading already makes it clear, an introductory paragraph that explains how this chapter contributes to the objectives of the report/project.

3.1 Heading level 2

3.1.1 Heading level 3

3.1.1.1 Deepest heading, only if you cannot do without it

Equations: An equation must read like part of the text. The solution of the quadratic equation $ax^2 + bx + c = 0$ given by the following expression (note the full stop after the equation to indicate the end of the sentence):

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2b}. \tag{3.1}$$

In other cases the equation is in the middle of the sentence. Then the paragraph following the equation should start with a small letter. Euler's identity is

$$e^{i\pi} + 1 = 0, \tag{3.2}$$

where e is Euler's number, the base of natural logarithms.

The `amsmath` has a wealth of structure and information on formatting of mathematical equations.

Symbols and numbers: Symbols that represent values of properties should be printed in italics, but SI units and names of functions (e.g. sin, cos and tan) must not be printed in italics. There must be a small hard space between a number and its unit, e.g. 120 km. Use the `siunitx` package to typeset numbers, angles and quantities with units:

`\num{1.23e3}` → 1.23×10^3
`\ang{30}` → 30°
`\qty{20}{N.m}` → 20 N·m

Figures and tables: The `graphicx` package can import `PDF`, `PNG` and `JPG` graphic files.

Table 3.1: Standard ISO paper sizes

Paper	Sizes	
	W [mm]	*H* [mm]
A0	841	1189
A1	594	841
A2	420	594
A3	297	420
A4	210	297
A5	148	210

Figure 3.1: Water plants

Chapter 4

Conclusions

Appendix A

Mathematical proofs

A.1 Euler's equation

Euler's equation gives the relationship between the trigonometric functions and the complex exponential function.

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{A.1}$$

Inserting $\theta = \pi$ in (A.1) results in Euler's identity

$$e^{i\pi} + 1 = 0 \tag{A.2}$$

A.2 Navier Stokes equation

The Navier–Stokes equations mathematically express momentum balance and conservation of mass for Newtonian fluids. Navier-Stokes equations using tensor notation:

$$\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial x_j}\left[\rho u_j\right] = 0 \tag{A.3a}$$

$$\frac{\partial}{\partial t}(\rho u_i) + \frac{\partial}{\partial x_j}\left[\rho u_i u_j + p\delta_{ij} - \tau_{ji}\right] = 0, \quad i = 1, 2, 3 \tag{A.3b}$$

$$\frac{\partial}{\partial t}(\rho e_0) + \frac{\partial}{\partial x_j}\left[\rho u_j e_0 + u_j p + q_j - u_i\tau_{ij}\right] = 0 \tag{A.3c}$$

Appendix B

Experimental results

List of references

[1] L. Engineering, “The advantages of using a portable cmm arm in metrology,” *Leussink Engineering*, 2024.

[2] S. Ibaraki and R. Saito, “Novel kinematic model of articulated arm coordinate measuring machine with angular position measuremnt errors of rotary axes,” *CIRP Annals*, vol. 72, no. 1, pp. 449–452, 2023.

[3] B. Alvarez, E. Cuesta, P. Luque, D. Mantaras, and D. Muina, “Dynamic deformations in coordinate measuring arms using virtual simulation,” *Internation Journal of Simulation Modelling*, vol. 14, no. 4, pp. 609–620, 2015.

[4] E. Cuesta, D. A. Mantaras, P. Luque, B. J. Alvarez, and D. Muina, “A calibration method of portable coordinate measuring arms by using artifacts,” *MAPAN-Journal of Metrology Society of India*, vol. 34, no. 1, pp. 1–11, 2019.

[5] L. Hall and K. Schreve, “Error distribution of an articulated arm measurement system,” (Stellenbosch, South Africa), July 2010.

[6] H. Meagher, “Measurement tool spotlight - portable measuring arms.” OASIS Alignment Service, LLC, February 2016.